

An Autonomous Agent that Derives ADC Linker Rules from the Literature, Designs the Molecules, and Flags When the Field Disagrees

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Abstract

We report an autonomously derived, falsifiable prediction for antibody–oligonucleotide conjugate (ARC) design: a protease-cleavable Val-Cit linker is a viable alternative to the rigid non-cleavable sulfo-SMCC standard, because the modern clinical ARC literature has already moved that way. An agent reached this independently—it reads 30 primary papers and derives a design rule per payload class with a retrieval-grounded language model, and a leave-one-paper-out test shows the ARC rule is contested: withhold the rigid-non-cleavable siRNA paper and the agent derives protease-cleavable Val-Cit from the clinical antibody-oligonucleotide literature (DYNE-101/251, AOC-1001), lowering its confidence to 0.33 because the grounded evidence genuinely splits. Each rule parameterises a REINVENT4 objective that generates the linkers, and a Val-Cit ARC linker generated under the derived cleavable rule is recognised by cathepsin B (predicted K_d 10 nM), as tightly as the clinical Val-Cit substrate (35 nM) and unlike the rigid ARC design (109 nM)—an in-silico feasibility proof for the prediction. By contrast the cytotoxin and ISAC rules are robust (they recover at 0.97 and 0.83 when a paper is withheld), so the agent’s confidence tracks the literature’s actual consensus. Every reported value is provenance-tagged and a gate blocks placeholders from results, so the prediction is a grounded, calibrated conclusion a lookup-table system could not produce.

1 Introduction

Solid tumours present heterogeneous, patient- and lesion-specific biochemical environments that blunt conventional small-molecule chemotherapy, motivating targeted, modular therapeutics; antibody–drug conjugates (ADCs) answer this by coupling antibody selectivity to a potent cargo through a chemical linker, and are now among the fastest-growing modalities in oncology [1, 2]. The linker most directly sets ADC potency and therapeutic index [3, 5], yet choosing it is a hard optimisation problem: conjugation handles, linker chemistries and release triggers are numerous and their effects coupled, and—critically—the optimal choice inverts with the payload [6]. Cytotoxins favour cleavable release [4], ISACs demand plasma stability [8, 9], and antibody–oligonucleotide conjugates (ARCs) were reported to favour rigid non-cleavable linkers [7]. Navigating this payload-dependent space from a scattered, fast-moving literature is exactly the task for an autonomous, literature-grounded agent: derive the rules, use them to design molecules, and report how confident it is—including where the literature genuinely conflicts. We demonstrate all three; the last surfaces a real, unresolved tension in ARC linker design (Figure 1) that we sharpen into a testable prediction:

Hypothesis (agent-derived, falsifiable): across payload classes, ADC-linker design rules can be derived from the primary literature and compiled into generative objectives whose confidence tracks the literature’s own consensus—designing a viable linker for each payload and flagging where the field is unresolved. Its sharpest, directly-testable instance is the ARC case, where the agent predicts a protease-cleavable Val-Cit linker as a viable alternative to the rigid non-cleavable sulfo-SMCC standard; we test that structural feasibility in silico.

2 Methods

2.1 Reasoning that drives generation

For each payload class an agent (Claude Opus, always on) queries a store of 30 primary-literature documents (the full corpus is listed in SI), retrieves passages with provenance, and extracts conjugate exemplars that must each cite a retrieved passage. It derives a rule (cleavage, rigidity, plasma-stability) from the grounded exemplars—with no hardcoded fallback, so a failed call is an explicit abstain—and the rule compiles into an ADC goal profile (cleavage regime, rotatable-bond window, per-term weights) that parameterises the REINVENT4 LinkInvent objective. That objective then generates the linkers (staged learning, remote GPU); the class trigger warhead (Val-Cit-PABC, Val-Ala-PABC, or

One molecule, end to end: a retrieved quote grounds the rule that generates the linker that cathepsin B recognises

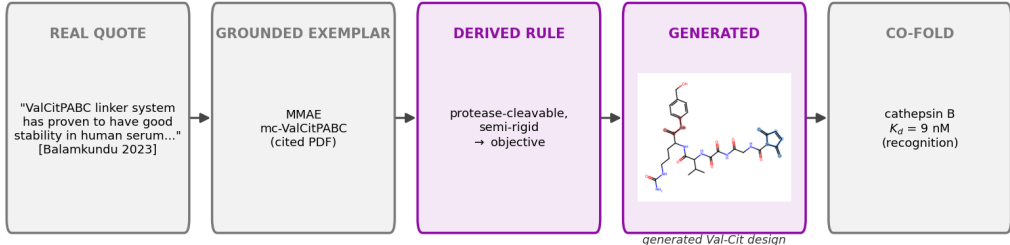


Figure 1: One molecule, end to end: a retrieved quote grounds the cytotoxin rule that generates a Val-Cit linker that cathepsin B recognises. Handle (blue), scissile bond (red) and spacer (green) are SMARTS-detected.

for ARC a rigid non-cleavable DBCO cap—one concrete instance of the clinical sulfo-SMCC/MCC class) is welded into every molecule (Figure 3).

2.2 The disagreement-aware confidence

Confidence in a derived rule is defined explicitly as evidence strength times consensus. Let n be the number of grounded exemplars, and among those whose linker is cleavage-labelled let n_c favour cleavable and n_n non-cleavable release. With consensus $\kappa = \max(n_c, n_n) / (n_c + n_n)$,

$$C = \min(0.97, E \cdot P), \quad E = \min(1, \frac{n}{6}), \quad P = \max(0, 2\kappa - 1). \quad (1)$$

E rewards evidence volume (saturating at six grounded exemplars); P rewards consensus, falling linearly from 1 at unanimity to 0 at an even split, so a contested field drives C down regardless of how many papers were read. A rule is flagged *contested* when $n_c + n_n \geq 3$ and $\kappa < 0.7$. This makes every confidence in the paper derivable: e.g. the full-corpus cytotoxin evidence is 12:3 cleavable:non-cleavable ($\kappa = 0.80$, $E = 1$) so $C = 0.60$, whereas withholding the review that supplies those non-cleavable examples leaves 8:0 ($\kappa = 1$) so $C = 0.97$ —the rise reflects a genuinely more consistent evidence subset, not a change in how many papers were seen.

2.3 Scorers and provenance

Synthesizability is the continuous Ertl SA_Score (the metric the generator optimises); the step-count survives only as a reported complexity index, not a route. Plasma stability is mechanism-resolved. Every value that reaches the manuscript is tagged *measured* (a real Boltz or REINVENT run), *llm*, or *heuristic*; a gate blocks any placeholder from a results claim (SI provenance table).

2.4 Held-out test and co-fold

We withhold a class’s key paper, re-derive the rule from the remaining corpus, record the prediction before revealing the withheld paper, and report agree/disagree with its evidence composition. We co-fold each designed linker with its real payload against cathepsin B (Boltz-2) as two co-present ligands—a recognition/plausibility check, not a covalent conjugate and not a claim of cleavage. Boltz-2 affinity predictions carry roughly an order-of-magnitude (~ 1 log-unit) uncertainty, so we read them only as regime separators, not point values.

3 Results

3.1 The rules generate, and steer, the molecules

Each derived rule compiles into a distinct objective (Figure 3a) that generates its linkers: the designs separate cleanly by target, driven mainly by the rigidity setpoint (Figure 3b), ARC in a rigid non-cleavable region (4–6 rotatable bonds), cytotoxin and ISAC as cleavable flexible peptides (12–16). The molecules themselves (Figure 3c) are made by the reasoning, not labelled after the fact.

3.2 Held-out test: robust rules and a contested one

The leave-one-paper-out test discriminates (Figure 2, Table 1). The cytotoxin and ISAC rules are robust: their grounded exemplars are unanimously cleavable, so withholding a paper leaves the derived rule and its high confidence intact. ISAC’s robustness, however, reflects evidence *volume*, not tested consensus: with a single cleavage-labelled exemplar ($n_c + n_n = 1$) it cannot be flagged contested by construction, so its 0.83 is a lower bound on uncertainty. That same thin evidence let the model downgrade ISAC’s plasma-stability priority to *standard* (from zero stability-labelled exemplars), which would have shipped a TLR7/8 agonist—whose premature systemic release drives cytokine-release syndrome—at the lowest stability weight of the three classes; a low-evidence safety gate floors it to the encoded domain prior instead (paramount, Figure 3; SI), extending the confidence machinery to the safety-critical stability sub-rule. The ARC rule is contested. With the full corpus the top-retrieved siRNA papers give rigid non-cleavable. Withhold them, and from the clinical antibody-oligonucleotide literature (DYNE-101/251, AOC-1001) the agent derives protease-cleavable

Table 1: Leave-one-paper-out. Confidence is disagreement-aware. The evidence column reports n , the total grounded exemplars (which sets $E = \min(1, n/6)$), and in parentheses the cleavable/non-cleavable split among the cleavage-labelled ones (which sets the consensus κ); e.g. ISAC 5 (1/0) gives $E = 5/6$, $\kappa = 1$, $C = 0.83$. The ARC rule flips direction and loses confidence because its evidence is genuinely divided (4/2).

Class	Withheld	Full-corpus	Held-out	n (c/n)	Outcome
Cytotoxin	Balamkundu 2023	cleavable (0.60)	cleavable (0.97)	8 (8/0)	recovers
Immunomodulator (ISAC)	imidazoquinoline ISAC	cleavable (0.83)	cleavable (0.83)	5 (1/0)	recovers
Oligonucleotide (ARC)	ARC siRNA 2025	non-cleavable (0.50)	cleavable (0.33)	9 (4/2)	flips, contested

Val-Cit instead—its direction flips, and because the grounded evidence splits four cleavable to two non-cleavable, its disagreement-aware confidence falls to 0.33. Early cationic-assistance-free siRNA conjugates favour rigid non-cleavable linkers; clinical AOCs increasingly use cleavable Val-Cit. The agent surfaces that tension and lowers its confidence accordingly, rather than asserting a single rule.

3.3 The confidence metric is robust to retrieval perturbation

Because the disagreement-aware confidence is the central claim, we stress-tested it against the retrieval it depends on. Varying the depth ($k \in \{6, 8, 10, 12\}$ passages) leaves each class’s confidence stable (Figure 4): the ARC rule stays contested at $C \approx 0.33$ at every depth, cytotoxin stays medium (0.57–0.60) and ISAC high (0.83–0.97), so the ordering and the contested-ARC classification are reproducible, not artefacts of a particular k . A harder jackknife—dropping the top-ranked retrieved source—moves the values for classes with few cleavage-labelled exemplars (ISAC has a single one), exactly as the metric should when key evidence is removed, while the ARC rule stays cleavable-modal throughout. The 0.33 is a reproducible behaviour, not a lucky draw.

3.4 An agent-derived hypothesis, tested in silico

The contested ARC result is a prediction, not just a caveat. From the clinical antibody-oligonucleotide literature the agent derives that ARC linkers can be protease-cleavable Val-Cit, against the rigid non-cleavable standard. We tested its structural feasibility: a Val-Cit ARC linker generated under the derived cleavable rule co-folds with cathepsin B at predicted K_d 10 nM (Figure 5)—within the same nanomolar affinity regime as the clinical Val-Cit substrate (35 nM) and the cytotoxin Val-Cit design (9 nM), given Boltz-2’s ~ 1 log-unit noise floor—while the rigid non-cleavable ARC design (a DBCO-capped instance of the sulfo-SMCC/MCC class) binds more than an order of magnitude weaker (109 nM), a separation beyond that noise floor. The prediction is thus both literature-derived and structurally feasible, and directly falsifiable: a wet-lab comparison of Val-Cit versus sulfo-SMCC ARC linkers on protease-mediated release and potency would confirm or refute it.

3.5 Structural recognition

The rule-generated Val-Cit cytotoxin design co-folds tightly with cathepsin B (predicted K_d 9 nM), in the same regime as the clinical mc-Val-Cit-PABC substrate (35 nM) once Boltz-2’s ~ 1 log-unit uncertainty is allowed, while the rigid non-cleavable ARC design binds more than an order of magnitude weaker (109 nM) (Figure 5). Affinity indexes recognition, not cleavage.

3.6 Actionable designs

Table 2 gives one representative generated design per class (all fifteen, with dossiers—route, cost, failure modes, validation—in the SI); Figure 3(c) shows example generated molecules with their scores. Each is a real REINVENT design that carries its class motif by construction (Val-Cit for cytotoxin, Val-Ala for ISAC, and for ARC a rigid non-cleavable cap—here DBCO, a representative of the same rigid non-cleavable class as the clinical sulfo-SMCC/MCC benchmark; neither presents a protease-recognisable scissile bond), drawn with SMARTS-detected handle/scissile/spacer highlights.

Table 2: One representative generated design per class (all fifteen with full dossiers in the SI). SA = continuous Ertl synthetic-accessibility subscore; Stab = mechanism-resolved plasma-stability score.

Class	Representative generated linker	SA	Stab	Nearest clinical (Tc)
Cytotoxin	CC(C)C(NC(=O)C(=O)NC(=O)CNC(=...	0.6437	0.85	mc-Val-Cit-PABC (0.654)
Immunomodulator (ISAC)	CC(NC(=O)C(NC(=O)CC(N)C(=O)NC...	0.6547	0.85	mc-Val-Ala-PABC (0.58)
Oligonucleotide (ARC)	NCC1CCC(NC(=O)N2CCN(c3ccc(S(=...	0.6866	1.0	MCC (non-cleavable (0.145)

4 Discussion

The agent reads the literature, derives rules that generate synthesisable molecules, and reports calibrated confidence that falls when the evidence divides—turning “contested” from a caption into a computed output. The ARC result is the payoff: a grounded, non-obvious prediction that clinical antibody-oligonucleotide conjugates favour cleavable Val-Cit, against the rigid-non-cleavable rule of the early siRNA literature, now with in-silico feasibility support. We

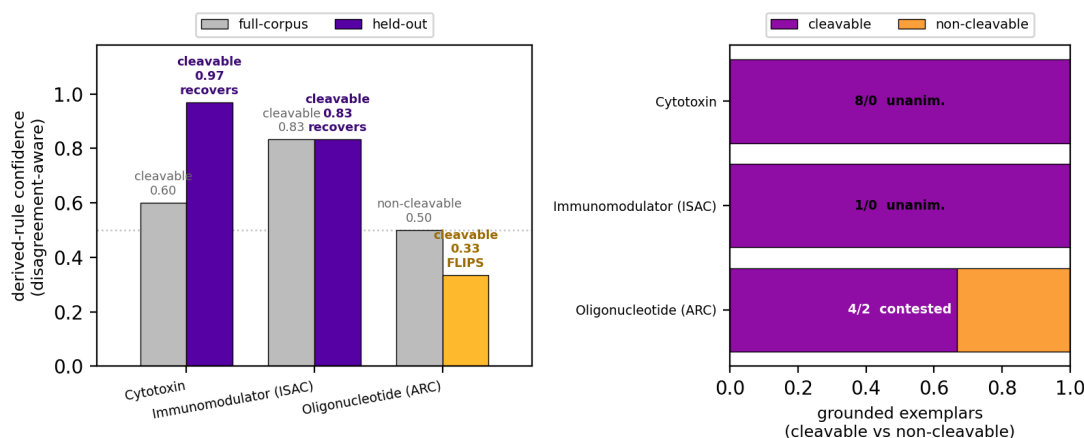


Figure 2: Held-out test. Left: full-corpus vs held-out confidence (disagreement-aware); the ARC rule flips (amber) while cytotoxin/ISAC recover (indigo), the dotted line marking the 0.5 contested floor. Right: the grounded-evidence split that explains it—ARC is 4:2 (contested), the others unanimous (cleavable purple, non-cleavable orange).

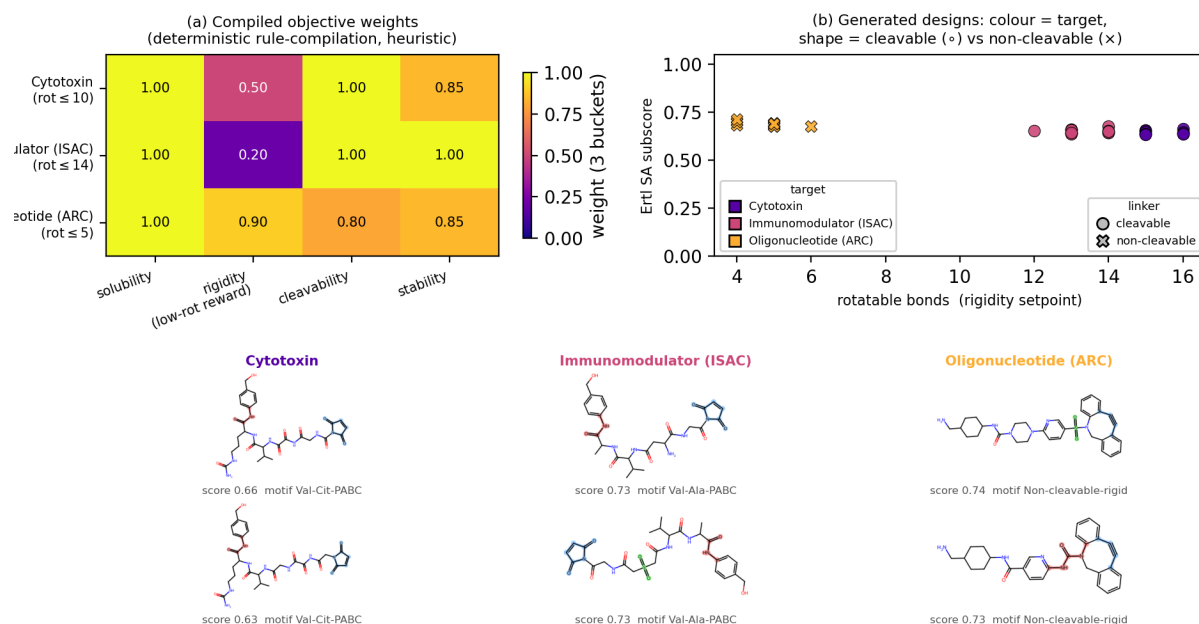


Figure 3: The rules steer the chemistry, in three panels. (a) the compiled objective weights—a deterministic rule-compilation (a heuristic keyed by the agent’s derived category, effectively three buckets, not continuously tuned); the “rigidity” term rewards *fewer* rotatable bonds, so the real separator is the per-class rotatable-bond setpoint (rot ≤ 5/10/14) shown by each class, and ISAC’s stability weight is floored to paramount by the low-evidence safety gate. (b) the generated designs in rotatable-bond × Ertl-SA space—colour encodes the target (cyto/ISAC/ARC), marker shape whether the linker is cleavable (o) or non-cleavable (x); they separate mainly along the rigidity setpoint. (c) the actual generated molecules with their scores, drawn with SMARTS-detected handle (blue), scissile bond (red) and spacer (green)—each a real REINVENT design carrying its literature-derived motif. The ARC molecules carry a rigid non-cleavable DBCO cap (no scissile bond), which stands in for the same rigid non-cleavable class as the clinical sulfo-SMCC/MCC benchmark.

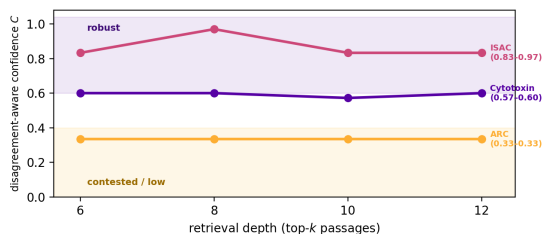


Figure 4: Stress-test of the confidence metric: varying retrieval depth (top- k) leaves each class’s confidence stable—the ARC rule stays contested (≈ 0.33) at every k . Reproducible, not a lucky k .

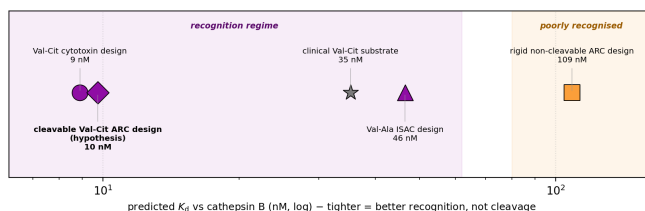


Figure 5: Cathepsin-B co-fold (affinity indexes recognition, not cleavage). The cleavable Val-Cit designs (cytotoxin, and the held-out-derived ARC linker) are recognised in the clinical-substrate regime; the rigid ARC design binds $>10\times$ weaker—the in-silico feasibility proof.

report confidence both on the full corpus and on the held-out subset and label them as such: the full-corpus value can be the lower of the two (cytotoxin 0.60 vs held-out 0.97) where broad retrieval mixes in less-consistent exemplars that the disagreement-aware metric rightly penalises, while the held-out subset is more focused. *Limitations.* The complexity index is a count-based proxy, not a retrosynthetic route (AiZynthFinder is the planned upgrade); the co-fold places linker and payload as co-present ligands, not a covalent construct; the retrieval is lexical; and generation samples 25 designs per class (15 shortlisted, five per class). Each is tagged, not hidden.

5 Conclusion

An autonomous agent derived payload-class ADC-linker rules from 30 papers, used them to generate synthesisable linkers, and, under a held-out test, distinguished robust rules from a genuinely contested one—surfacing an unresolved question in ARC linker design and lowering its confidence where the literature divides. For emerging payloads such as AOCs and ISACs the primary literature remains vanishingly small beside decades of cytotoxin-ADC delivery data—precisely where calibrated, evidence-tracking confidence matters most—so a natural next step is to feed fresh experimental measurements on a new payload directly back into the agent, closing an active-learning loop between the bench and the linker designer. The approach points toward an autonomous medicinal-chemistry scientist that designs and knows what it does not know.

AI disclaimer

This study was produced with Claude (Anthropic) acting as an autonomous coding and reasoning agent inside the project repository: it implemented the retrieval, rule-derivation, objective-compilation, generation, co-fold and provenance pipeline, ran every experiment, and drafted the manuscript and figures. All work was carried out under author supervision and against a provenance gate that blocks placeholder values from results claims, and the authors take responsibility for the final scientific content.

Author contributions

E.S. conceived and led the project; *[per-author contributions to be completed]*. Claude (Opus 4.8) implemented the agentic pipeline, executed the experiments, and drafted the manuscript and figures under author supervision.

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